
Configuring and Deploying a Relational Database Environment

This lab focuses on setting up and configuring a database, building upon the concepts explored in the relational database module (S4). Topics include tablespaces, user creation, Data Definition Language (DDL) commands like CREATE TABLE, ALTER TABLE, DROP TABLE, and Data Manipulation Language (DML) commands such as SELECT, INSERT, UPDATE, DELETE,...

① Useful Notions & Syntaxes

- (a) **Tablespace Creation.** The fundamental syntax is:

```
CREATE TABLESPACE TABLESPACE_Name  
DATAFILE 'C:\path\TABLESPACE_Name.dat'  
SIZE 100M  
AUTOEXTEND ON  
ONLINE;
```

- (b) **Temporary Tablespace Creation.** The basic syntax is:

```
CREATE TEMPORARY TABLESPACE TempTABLESPACE_Name  
TEMPFILE 'C:\path\TempTABLESPACE_Name.dat'  
SIZE 100M  
AUTOEXTEND ON;
```

- (c) **User Creation.** The syntax framework is:

```
alter session set "_ORACLE_SCRIPT"=true;  
  
CREATE USER UserName  
IDENTIFIED BY PasseWord  
DEFAULT TABLESPACE TABLESPACE_Name  
TEMPORARY TABLESPACE TempTABLESPACE_Name;
```

- (d) **Grant All Rights to the User, and Connection.** The basic syntax is:

```
GRANT ALL PRIVILEGES TO UserName;
connect UserName/PassWord@XE;
```

- (e) **Table Creation, Constraints, and Other Manipulations.** As the following syntax:

```
CREATE TABLE table_name (
    column1 datatype1 [NOT NULL],
    column2 datatype2 [NOT NULL],
    CONSTRAINT constraintName1 constraintType));
...
);
----- More Info about constraints
Four types of constraints
-UNIQUE (column1[,column2])
-PRIMARY KEY (column1[,column2])
-FOREIGN KEY (column1 [,column2]) REFERENCES [schma.]parentTableName
(column1 [,column2]) [ON DELETE { CASCADE | SET NULL }]
-CHECK (condition)

--Deleting a Table
DROP TABLE Table_Name;

ALTER TABLE table_name
ADD CONSTRAINT constraint_name constraint_type (column_name);
```

- (f) **Data Manipulation Language** . As following:

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);

SELECT column_name FROM table_name;

UPDATE table_name
SET column_name = new_value
WHERE condition;

DELETE FROM table_name
WHERE condition;
```

② Lab Work

We will work with an investigative journalist in the world of cinema, and we will employ database tools and techniques to interrogate the data, discover key patterns, and drawing conclusions. Get ready to query, analyze, and report on the secrets of the cinematic universe!

- i) **PreLab:** Download Oracle from the link: <https://bit.ly/49kjNkL>, **Before the Lab Session;**
- ii) IMDb (Internet Movie Database) is a sample database that contains movie-related information, with around (1050000) records for each table. Check the database at:
 - 1- <https://imdb.to/49kjM0h> ;
 - 2- <https://datasets.imdbws.com/>;
 - 3- Download the filtrated db and its schema: <https://drive.google.com/drive/folders/1sJy5V5h49WFLsYCZOzRnpj8mBT04l1NV?usp=sharing>

- iii) **Discuss the schema of the db.**
- iv) Create two TableSpaces "imijdb_TBS" and "imijdb_TempTBS" of 1 Go.
- v) Create a user "imijdbManager" with password "imijdb25" by assigning it the two tablespaces created, and Grant all privileges to "imijdbManager" .**Verify the creation of the user.**
- vi) Create tables discussed in (Q2). **Verify that the tables were created correctly.**
- vii) Perform the following SQL queries:
 - Insertion of 20 rows in each table;
 - Selection of everything in table "ratings", where average rating equals "6";
 - Find the movies released in a specific year;
 - Retrieve information about a specific movie;
 - Find the average rating for a particular genre, like "Action";
 - Retrieve the top 10 movies by rating;
 - Create a *Trigger* to *Update* "endYear" of "Show" When a New Episode is Added;
 - Create a *Trigger* to Automatically Insert a Rating Record (Default Values e.g., aveRating 0, numVotes 0) When a New Show is Added

③ Homework

By the end of each Lab, a report should be done, that contains all the details of the elaborated work during the Lab session. The report is going to be submitted later on. **You can work in groups of three maximum.**